

CS 562: Artificial Intelligence

Course Code	CS 562
Course Name	Artificial Intelligence
L-T-P-C	3-0-0-3
Prerequisites	IC250 - Programming and Data Structure Practicum or Equivalent, CS202 Data Structure and Algorithm or Equivalent, CS403 - Algorithm Design and Analysis or Equivalent, COT
Intended for	B. Tech, MS, M. Tech. & PhD.
Distribution	Elective for Third and Final year B. Tech (All branches), MS, M. Tech. & PhD
Approval	18th BoA (Sep 2017); 14th Senate (Oct 2017)

Course Contents

- **Introduction:** Overview and Historical Perspective, Turing Test, Physical Symbol Systems and the scope of Symbolic AI, Agents. [3 Lectures]
- **State Space Search:** Depth First Search, Breadth First Search, DFID. [3 Lectures]
- **Heuristic Search:** Best First Search, Hill Climbing, Beam Search. [3 Lectures]
- Traveling Salesman Problem, Tabu Search, Simulated Annealing. [3 Lectures]
- **Population Based Search:** Genetic Algorithms, Ant Colony Optimization. [3 Lectures]
- Branch & Bound, Algorithm A*, Admissibility of A*. [3 Lectures]
- Monotone Condition, IDA*, RBFS, Pruning OPEN and CLOSED in A* [3 Lectures]
- Problem Decomposition, Algorithm AO*, Game Playing. [3 Lectures]
- **Game Playing:** Algorithms Minimax, AlphaBeta, SSS*. [3 Lectures]
- Rule Based Expert Systems, Inference Engine, Rete Algorithm. [3 Lectures]
- **Planning:** Forward/Backward Search, Goal Stack Planning, Sussmans Anomaly. [3 Lectures]
- Plan Space Planning, Algorithm Graph plan. [3 Lectures]
- Constraint Satisfaction Problems, Algorithm AC-1, Knowledge Based Systems. [3 Lectures]
- Propositional Logic, Resolution Refutation Method. [3 Lectures]

Text Books

1. Deepak Khemani, **A First Course in Artificial Intelligence**, McGraw Hill Education, 2013.

References

1. John Haugeland, **Artificial Intelligence: The Very Idea**, A Bradford Book, The MIT Press, 1985.
2. Pamela McCorduck, **Machines Who Think: A Personal Inquiry into the History and Prospects of Artificial Intelligence**, 2nd Edition, A K Peters/CRC Press;2004.
3. Eugene Charniak and Drew McDermott, **Introduction to Artificial Intelligence**, Addison Wesley, 1985.
4. Zbigniew Michalewicz and David B. Fogel, **How to Solve It: Modern Heuristics**, 2nd Edition, Springer, 2004.
5. Judea Pearl, **Heuristics: Intelligent Search Strategies for Computer Problem Solving**, Addison-Wesley, 1984.
6. Elaine Rich and Kevin Knight, **Artificial Intelligence**, Tata McGraw Hill, 1991.
7. Stuart Russell and Peter Norvig, **Artificial Intelligence: A Modern Approach**, 3rd Edition, Prentice Hall, 2009.
8. Patrick Henry Winston, **Artificial Intelligence**, Addison-Wesley, 1992.
9. Stefan Edelkamp and Stefan Schroedl, **Heuristic Search: Theory and Applications**, Morgan Kaufmann, 2011.